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Model IV (M/M/S) : (N/FCFS)

This model is essentially the same as model III, except that the maximum number of customers in the system is limited to N, where $N > S$ (S = No. of channels)

$$\therefore \quad \lambda_n = \begin{cases} \lambda, & 0 \le n \le N \\ 0, & n \ge N \end{cases}$$

$$\mu_n = \begin{cases} n\mu, & 0 \le n \le S \\ S\mu, & S \le n \le N \end{cases}$$

Formula:

1) $$P_n = \begin{cases} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n P_0, & 0 \le n \le S \\ \frac{1}{S^{n-S}\, S!}\left(\frac{\lambda}{\mu}\right)^n P_0, & S \le n \le N \end{cases}$$

2) $$P_0 = \left[\sum_{n=0}^{S-1} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n + \sum_{n=S}^{N} \frac{1}{S^{n-S} S!} \left(\frac{\lambda}{\mu}\right)^n \right]^{-1}$$

3) $$L_q = \sum_{n=S}^{N} (n-S) P_n$$

4) $$L_s = L_q + S - P_0 \sum_{n=0}^{S-1} \frac{(S-n)(Sf)^n}{n!}$$

5) $$W_q = W_s - \frac{1}{\mu}$$

6) $$W_s = \frac{L_s}{\lambda(1-P_N)}$$

7) Traffic intensity, $f = \frac{\lambda}{\mu S}$ or, $Sf = \frac{\lambda}{\mu}$

Numerical Problems on (M/M/S) : (N/FCFS)

Q: A barber shop has two barber and three chairs for customers. Assume that the customers arrive in a poisson fashion at a rate of five per hour and each barber services customers according to an exponential distribution with 15 minutes. Further, if a customer arrives and there are no empty chairs in the shop, he will leave. What is the expected no. of customers in the shop?

Solution:

$S = 2$

$N = 3$

$\lambda = \frac{5}{60} = \frac{1}{12}$ per min

$\mu = \frac{1}{15}$ per min,

$\frac{\lambda}{\mu} = \frac{1}{12} \times 15 = \frac{5}{4}$

$\rho = \frac{\lambda}{\mu S}$

$S\rho = \frac{\lambda}{\mu} = \frac{5}{4}$

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$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n + \sum_{n=s}^{N} \frac{1}{s^{n-s}\, s!} \left(\frac{\lambda}{\mu}\right)^n \right]^{-1}$$

$$= \left[\sum_{n=0}^{1} \frac{1}{n!} \left(\frac{5}{4}\right)^n + \sum_{n=2}^{3} \frac{1}{2^{n-2}\, 2!} \left(\frac{5}{4}\right)^n \right]^{-1}$$

$0! = 1$

$a^0 = 1$

$$= \left[\left(1 + \frac{5}{4}\right) + \left(\frac{1}{2} \times \frac{25}{16} + \frac{1}{2 \times 2} \left(\frac{125}{64}\right)\right) \right]^{-1}$$

$$= \left[1 + \frac{5}{4} + \frac{25}{32} + \frac{125}{256} \right]^{-1}$$

$$= \left[\frac{256 + 320 + 200 + 125}{256} \right]^{-1}$$

$$= \left[\frac{901}{256} \right]^{-1} = \frac{256}{901} = 0.28$$

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$$P_n = \begin{cases} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n P_0 , & 0 \le n < s \\ \frac{1}{s^{n-s} s!}\left(\frac{\lambda}{\mu}\right)^n P_0 , & s \le n \le N \end{cases}$$

if $n=3$,

$0 \le 3 < 2$ ×

$2 \le 3 \le 3$ ✓

$$= \begin{cases} \frac{1}{n!}\left(\frac{5}{4}\right)^n \times 0.28 , & 0 \le n < s \\ \frac{1}{2^{n-2} \times 2!}\left(\frac{5}{4}\right)^n \times 0.28 , & s \le n \le N \end{cases}$$

Thus, $L_s = L_q + s - P_0 \sum_{n=0}^{s-1} \frac{(s-n)(sf)^n}{n!}$

$$= \sum_{n=s}^{N} (n-s) P_n + s - P_0 \sum_{n=0}^{s-1} \frac{(s-n)(sf)^n}{n!}$$

$$= \sum_{n=2}^{3} (n-2)P_n + 2 - 0.28 \sum_{n=0}^{1} \frac{(2-n)\left(\frac{5}{4}\right)^n}{n!}$$

$$= 0 + P_3 + 2 - 0.28\left(2 + 1.25\right)$$

$$= P_3 + 2 - 0.28 \times 3.25$$

$$= P_3 + 1.09$$

$$= \left[\frac{1}{2^{3-2} \times 2!}\left(\frac{5}{4}\right)^3 \times 0.28\right] + 1.09$$

$$= \left[\frac{1}{4} \times (1.25)^3 \times 0.28\right] + 1.09$$

$$= \frac{1.9531 \times 0.28}{4} + 1.09 \qquad = \frac{0.5468}{4} + 1.09$$

$$= 1.226$$

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Numerical Problem 5 – Model I (M/M/1) : (∞/FCFS)

Q: In a public telephone booth, the arrivals on an average are 15 per hour. A call on an average takes three minutes. If there is just one phone, find

(i) The expected no. of callers in the booth at any time.

(ii) The proportion of the time, the booth is expected to be idle ?

Solution : $\lambda = 15$ per hour

$\mu = \frac{1}{3}$ per min. $= \frac{1}{3} \times 60 = 20$ per hour.

(i) Expected length of non-empty queue, $= \frac{\mu}{\mu - \lambda} = \frac{20}{20-15} = \frac{20}{5} = 4$

(ii) The service is busy $= \frac{\lambda}{\mu} = \frac{15}{20} = \frac{3}{4}$

$\therefore$ The booth is expected to be idle for $1 - \frac{3}{4} = \frac{1}{4}$ hour

$= 15$ min.

MODEL II (M/M/I) : (N/FCFS)

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This model differs from model I in the sense that, the maximum number of customers in the system is limited to N. Arrivals will not exceed N in any case.

The various measures of this model are,

1. $P_0 = \frac{1-\rho}{1-\rho^{N+1}}$ where, $\rho = \frac{\lambda}{\mu}$ ($\frac{\lambda}{\mu} > 1$ is allowed)

2. $P_N = \frac{1-\rho}{1-\rho^{N+1}} \rho^n$, for $n = 0, 1, 2, \ldots N$

3. $L_q = L_s - \frac{\lambda}{\mu}$

4. $L_s = P_0 \sum_{n=0}^{N} n \rho^n$

5. $L_q = L_s - \frac{\lambda}{\mu}$

6. $W_s = \frac{L_s}{\lambda}$

7. $W_q = W_s - \frac{1}{\mu}$

Q: In a railway marshalling yard, goods train arrive at the rate of 30 trains per day. Assume that the inter-arrival time follows an exponential distribution and the service time is also to be assumed as exponential with mean of 36 minutes. Calculate,

(i) the probability that the yard is empty.

(ii) the average queue length, assuming that the line capacity of the yard is nine trains.

(Considered $N = 9$).

Solution:

$$\lambda = \frac{30}{24 \times 60} = \frac{1}{48} \text{ train per min.}$$

$$\mu = \frac{1}{36} \text{ per min.}$$

$$\therefore \rho = \frac{\lambda}{\mu} = \frac{36}{48} = 0.75$$

(i) The probability that the queue is empty is given by,

$$P_0 = \frac{1-\rho}{1-\rho^{N+1}} \quad (\text{Where, } N=9)$$

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$$= \frac{1 - 0.75}{1 - (0.75)^{10}}$$

$$= \frac{0.25}{0.943} = 0.265$$

(ii) Average no. of train in the system,

$$L_s = P_0 \sum_{n=0}^{N} n\rho^n$$

$$= 0.265 \left[0 + \rho + 2\rho^2 + 3\rho^3 + 4\rho^4 + 5\rho^5 + 6\rho^6 + 7\rho^7 + 8\rho^8 + 9\rho^9 \right]$$

$$= (0.265) \left[0.75 + 1.125 + 1.265 + 1.2656 + 1.1865 + 1.0678 + 0.9343 + 0.80090 + 0.6757 \right]$$

$$= (0.265)(9.0702) = 2.5529 .$$

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Average queue length is given by,

$$L_q = L_s - \frac{\lambda}{\mu}$$

$$= (2.5529 - 0.75)$$

$$= 1.8029$$

$$= 2 \text{ (approx)}$$

MODEL - II (M/M/1) : (N/FCFS) Numerical - 02

Q: A barber shop has space to accommadate only 10 customer. He can serve only one person at a time. If a customer comes to his shop and finds it full, he goes to the next shop. Customer randomly arrive at an average rate $\lambda = 10$ per hour and the barber's service time is negative exponential with an average of $1/\mu = 5$ minutes per customer. Find P_0 and P_n.

Solution :

$$N = 10$$

$$\lambda = \frac{10}{60} = \frac{1}{6} \text{ per min.}$$

$$\mu = \frac{1}{5} \text{ per min.}$$

$$\rho = \frac{\lambda}{\mu} = \frac{\frac{1}{6}}{\frac{1}{5}} = \frac{5}{6} = 0.8334$$

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(i) $P_0 = \frac{1-\rho}{1-\rho^{N+1}}$

$= \frac{1-0.8334}{1-(0.8334)^{11}} = \frac{0.1667}{0.8654}$

$= 0.1925$

(ii) $P_n = \frac{1-\rho}{1-\rho^{N+1}} \rho^n$

$= P_0 \rho^n$

$= (0.1925)(0.8334)^n \quad [n = 0, 1, 2, \ldots 10]$